

introduction to opencv & image processing

In [1]: `pip install opencv-python`

Collecting opencv-python

Downloading opencv_python-4.13.0.92-cp37-abi3-win_amd64.whl.metadata (20 kB)

Requirement already satisfied: numpy>=2 in c:\users\abhilash reddy\anaconda3\lib\site-packages (from opencv-python) (2.3.5)

Downloading opencv_python-4.13.0.92-cp37-abi3-win_amd64.whl (40.2 MB)

```

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```

Installing collected packages: opencv-python

Successfully installed opencv-python-4.13.0.92

Note: you may need to restart the kernel to use updated packages.

In [2]: `import numpy as np`
`import matplotlib.pyplot as plt`
`%matplotlib inline`

In [3]: `import cv2`

In [4]: `image = cv2.imread(r'C:\Users\Abhilash Reddy\Desktop\goku.jpg')`

In [5]: `image`

```

Out[5]: array([[ 59,  20,  12],
               [ 60,  21,  13],
               [ 59,  20,  12],
               ...,
               [190, 118,  54],
               [191, 119,  55],
               [190, 119,  52]],

          [[ 58,  19,  11],
           [ 59,  20,  12],
           [ 58,  19,  11],
           ...,
           [191, 120,  53],
           [191, 120,  53],
           [191, 120,  53]],

          [[ 58,  19,  11],
           [ 58,  19,  11],
           [ 57,  18,  10],
           ...,
           [192, 121,  53],
           [192, 121,  53],
           [192, 121,  53]],

          ...,

          [[  0,  0,  0],
           [  0,  0,  0],
           [  0,  0,  0],
           ...,
           [  0,  0,  0],
           [  0,  0,  0],
           [  0,  0,  0]],

          [[  0,  0,  0],
           [  0,  0,  0],
           [  0,  0,  0],
           ...,
           [  0,  0,  0],
           [  0,  0,  0],
           [  0,  0,  0]],

          [[  0,  0,  0],
           [  0,  0,  0],
           [  0,  0,  0],
           ...,
           [  0,  0,  0],
           [  0,  0,  0],
           [  0,  0,  0]]], shape=(377, 745, 3), dtype=uint8)

```

```
In [6]: type(image)
```

```
Out[6]: numpy.ndarray
```

```
In [7]: image.shape
```

```
Out[7]: (377, 745, 3)
```

```
In [8]: plt.imshow(image)
```

```
Out[8]: <matplotlib.image.AxesImage at 0x14a38bae900>
```



but the above is of different colour when compared to original photo # now to get original image

```
In [15]: fix_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB) # TO GET ORIGINAL PHOTO WE ARE CONV  
fix_image
```

```

Out[15]: array([[ 12,  20,  59],
                [ 13,  21,  60],
                [ 12,  20,  59],
                ...,
                [ 54, 118, 190],
                [ 55, 119, 191],
                [ 52, 119, 190]],

                [[ 11,  19,  58],
                [ 12,  20,  59],
                [ 11,  19,  58],
                ...,
                [ 53, 120, 191],
                [ 53, 120, 191],
                [ 53, 120, 191]],

                [[ 11,  19,  58],
                [ 11,  19,  58],
                [ 10,  18,  57],
                ...,
                [ 53, 121, 192],
                [ 53, 121, 192],
                [ 53, 121, 192]],

                ...,

                [[  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0],
                ...,
                [  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0]],

                [[  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0],
                ...,
                [  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0]],

                [[  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0],
                ...,
                [  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0]]], shape=(377, 745, 3), dtype=uint8)

```

```
In [17]: fix_image.shape # we get the same answer as above
```

```
Out[17]: (377, 745, 3)
```

```
In [18]: plt.imshow(fix_image)
```

Out[18]: <matplotlib.image.AxesImage at 0x14a3a392fd0>



In [22]: `image_gray = cv2.imread(r'C:\Users\Abhilash Reddy\Desktop\goku.jpg', cv2.IMREAD_GRA`

In [23]: `image_gray.shape`

Out[23]: (377, 745)

In [24]: `image_gray`

Out[24]: `array([[22, 23, 22, ..., 107, 108, 107],
 [21, 22, 21, ..., 108, 108, 108],
 [21, 21, 20, ..., 109, 109, 109],
 ...,
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0]], shape=(377, 745), dtype=uint8)`

In [28]: `image_gray.min()`

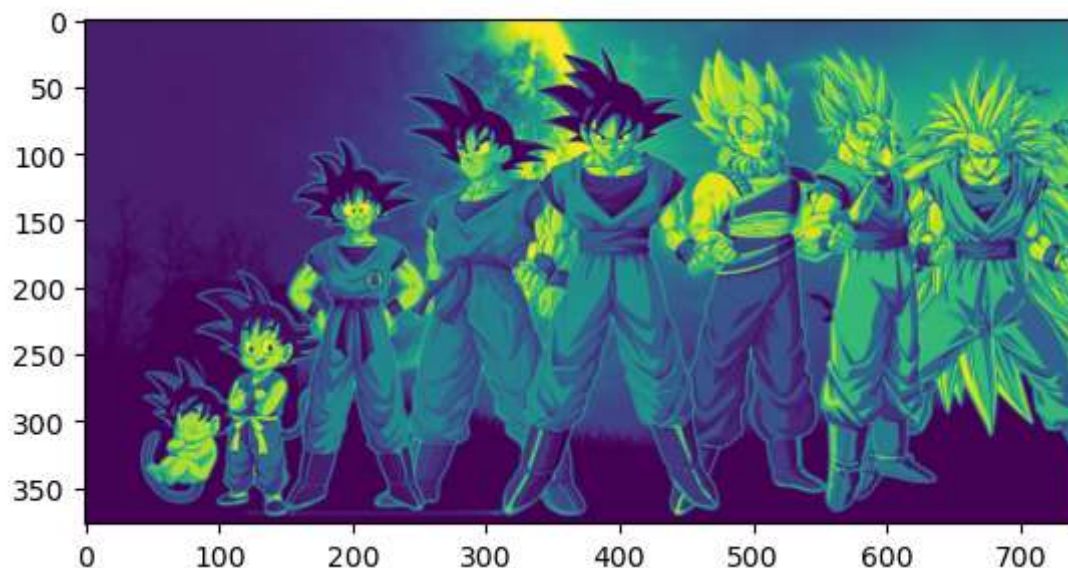
Out[28]: `np.uint8(0)`

In [29]: `image_gray.max()`

Out[29]: `np.uint8(255)`

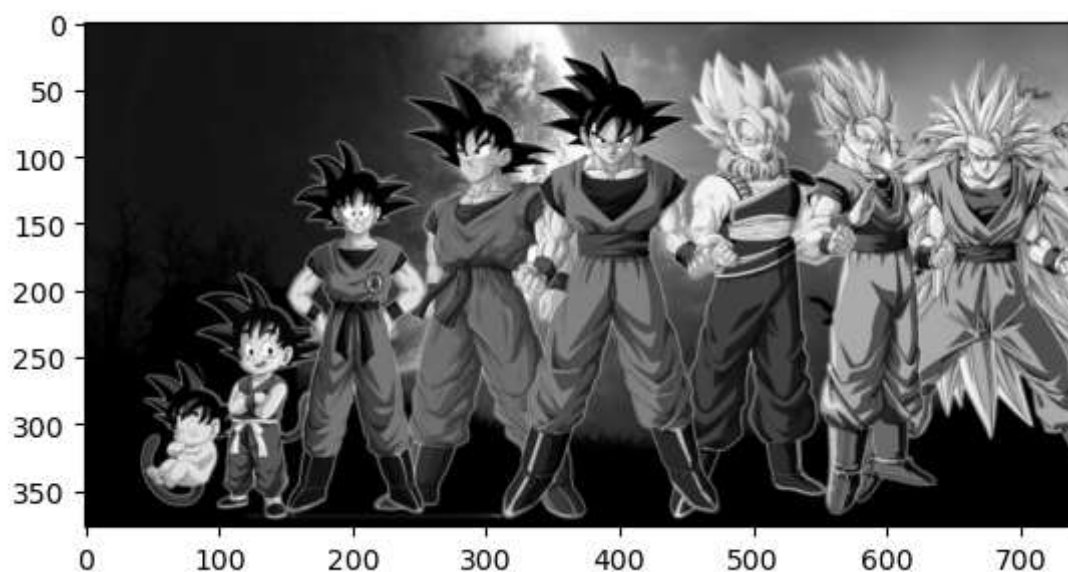
In [31]: `plt.imshow(image_gray)`

Out[31]: <matplotlib.image.AxesImage at 0x14a3bcaad50>



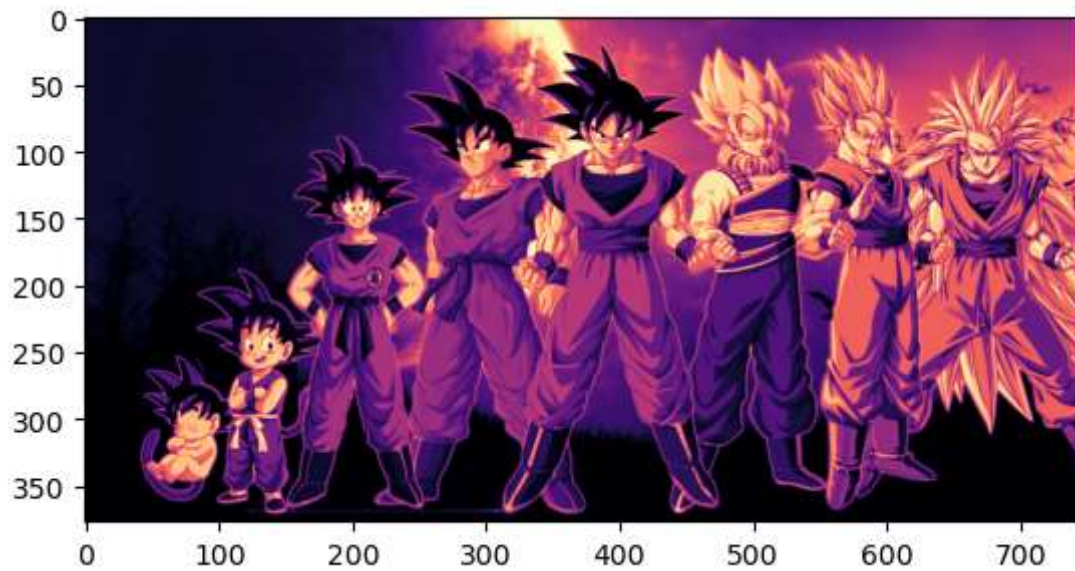
```
In [32]: plt.imshow(image_gray, cmap = 'gray')
```

```
Out[32]: <matplotlib.image.AxesImage at 0x14a3bc82c10>
```



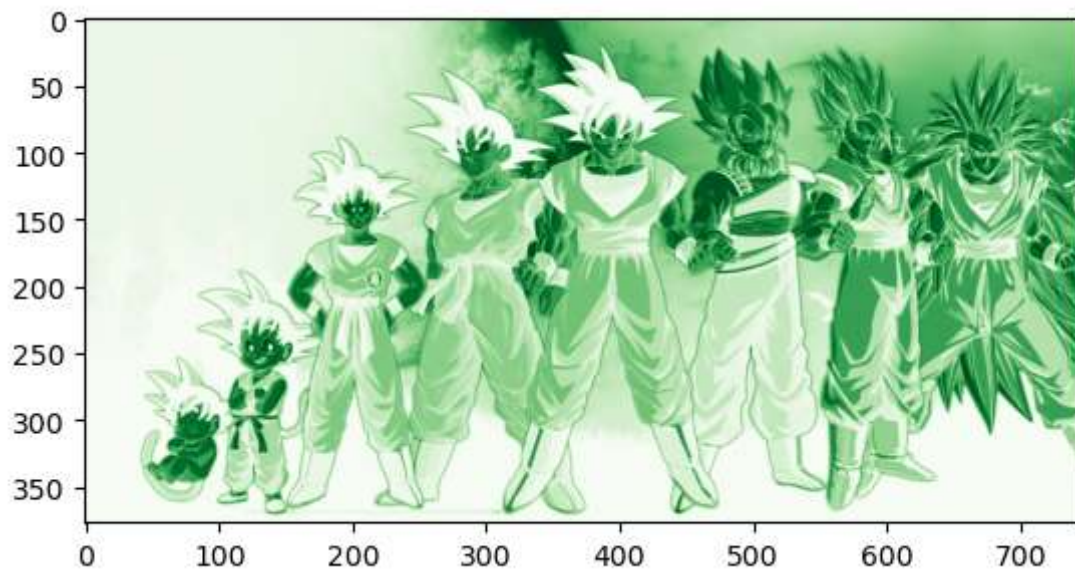
```
In [33]: plt.imshow(image_gray, cmap = 'magma')
```

```
Out[33]: <matplotlib.image.AxesImage at 0x14a3a866ad0>
```



```
In [34]: plt.imshow(image_gray, cmap = 'Greens')
```

```
Out[34]: <matplotlib.image.AxesImage at 0x14a3a8ba710>
```



```
In [36]: plt.imshow(image_gray, cmap = 'greens')
```

```

-----
ValueError                                Traceback (most recent call last)
Cell In[36], line 1
----> 1 plt.imshow(image_gray, cmap = )

File ~\anaconda3\Lib\site-packages\matplotlib\pyplot.py:3601, in imshow(X, cmap, norm, aspect, interpolation, alpha, vmin, vmax, colorizer, origin, extent, interpolation_stage, filternorm, filterrad, resample, url, data, **kwargs)
    3579 @_copy_docstring_and_deprecators(Axes.imshow)
    3580 def imshow(
    3581     X: ArrayLike | PIL.Image.Image,
    (...) 3599     **kwargs,
    3600 ) -> AxesImage:
-> 3601     __ret = gca().imshow(
    3602         X,
    3603         cmap=cmap,
    3604         norm=norm,
    3605         aspect=aspect,
    3606         interpolation=interpolation,
    3607         alpha=alpha,
    3608         vmin=vmin,
    3609         vmax=vmax,
    3610         colorizer=colorizer,
    3611         origin=origin,
    3612         extent=extent,
    3613         interpolation_stage=interpolation_stage,
    3614         filternorm=filternorm,
    3615         filterrad=filterrad,
    3616         resample=resample,
    3617         url=url,
    3618         **({ : data} if data is not None else {}),
    3619         **kwargs,
    3620     )
    3621     sci(__ret)
    3622     return __ret

File ~\anaconda3\Lib\site-packages\matplotlib\__init__.py:1524, in _preprocess_data.<locals>.inner(ax, data, *args, **kwargs)
    1521 @functools.wraps(func)
    1522 def inner(ax, *args, data=None, **kwargs):
    1523     if data is None:
-> 1524     return func(
    1525         ax,
    1526         *map(cbook.sanitize_sequence, args),
    1527         **{k: cbook.sanitize_sequence(v) for k, v in kwargs.items()})
    1529     bound = new_sig.bind(ax, *args, **kwargs)
    1530     auto_label = (bound.arguments.get(label_namer)
    1531                  or bound.kwargs.get(label_namer))

File ~\anaconda3\Lib\site-packages\matplotlib\axes\_axes.py:5968, in Axes.imshow(self, X, cmap, norm, aspect, interpolation, alpha, vmin, vmax, colorizer, origin, extent, interpolation_stage, filternorm, filterrad, resample, url, **kwargs)
    5750 @_preprocess_data()
    5751 @_docstring.interpd
    5752 def imshow(self, X, cmap=None, norm=None, *, aspect=None,
    (...) 5755     interpolation_stage=None, filternorm=True, filterrad=4.0,

```

```

5756         resample=None, url=None, **kwargs):
5757         """
5758         Display data as an image, i.e., on a 2D regular raster.
5759
5760         (...) 5966         (unassociated) alpha representation.
5761         5967         """
-> 5968         im = mimage.AxesImage(self, cmap=cmap, norm=norm, colorizer=colorizer,
5969                                interpolation=interpolation, origin=origin,
5970                                extent=extent, filternorm=filternorm,
5971                                filterrad=filterrad, resample=resample,
5972                                interpolation_stage=interpolation_stage,
5973                                **kwargs)
5975         if aspect is None and not (
5976             im.is_transform_set()
5977             and not im.get_transform().contains_branch(self.transData)):
5978             aspect = mpl.rcParams['image.aspect']

File ~\anaconda3\Lib\site-packages\matplotlib\image.py:884, in AxesImage.__init__(self, ax, cmap, norm, colorizer, interpolation, origin, extent, filternorm, filterrad, resample, interpolation_stage, **kwargs)
    867 def __init__(self, ax,
    868                *,
    869                cmap=None,
    (...) 879                **kwargs
    880                ):
    881         self._extent = extent
--> 884         super().__init__(
    885             ax,
    886             cmap=cmap,
    887             norm=norm,
    888             colorizer=colorizer,
    889             interpolation=interpolation,
    890             origin=origin,
    891             filternorm=filternorm,
    892             filterrad=filterrad,
    893             resample=resample,
    894             interpolation_stage=interpolation_stage,
    895             **kwargs
    896         )

File ~\anaconda3\Lib\site-packages\matplotlib\image.py:263, in _ImageBase.__init__(self, ax, cmap, norm, colorizer, interpolation, origin, filternorm, filterrad, resample, interpolation_stage, **kwargs)
    250 def __init__(self, ax,
    251                cmap=None,
    252                norm=None,
    (...) 261                **kwargs
    262                ):
--> 263         super().__init__(self._get_colorizer(cmap, norm, colorizer))
    264         if origin is None:
    265             origin = mpl.rcParams['image.origin']

File ~\anaconda3\Lib\site-packages\matplotlib\colorizer.py:610, in _ScalarMappable._get_colorizer(cmap, norm, colorizer)
    606         _ScalarMappable._check_exclusionary_keywords(
    607             Colorizer, cmap=cmap, norm=norm

```

```

608     )
609     return colorizer
--> 610 return Colorizer(cmap, norm)

```

File ~\anaconda3\Lib\site-packages\matplotlib\colorizer.py:56, in Colorizer.__init__(self, cmap, norm)

```

53 def __init__(self, cmap=None, norm=None):
55     self._cmap = None
--> 56     self._set_cmap(cmap)
58     self._id_norm = None
59     self._norm = None

```

File ~\anaconda3\Lib\site-packages\matplotlib\colorizer.py:237, in Colorizer._set_cmap(self, cmap)

```

235 from matplotlib import cm
236 in_init = self._cmap is None
--> 237 self._cmap = cm._ensure_cmap(cmap)
238 if not in_init:
239     self.changed()

```

File ~\anaconda3\Lib\site-packages\matplotlib\cm.py:309, in _ensure_cmap(cmap)

```

306 # use check_in_list to ensure type stability of the exception raised by
307 # the internal usage of this (ValueError vs KeyError)
308 if cmap_name not in _colormaps:
--> 309     _api.check_in_list(sorted(_colormaps), cmap=cmap_name)
310 return mpl.colormaps[cmap_name]

```

File ~\anaconda3\Lib\site-packages\matplotlib_api__init__.py:130, in check_in_list(values, _print_supported_values, **kwargs)

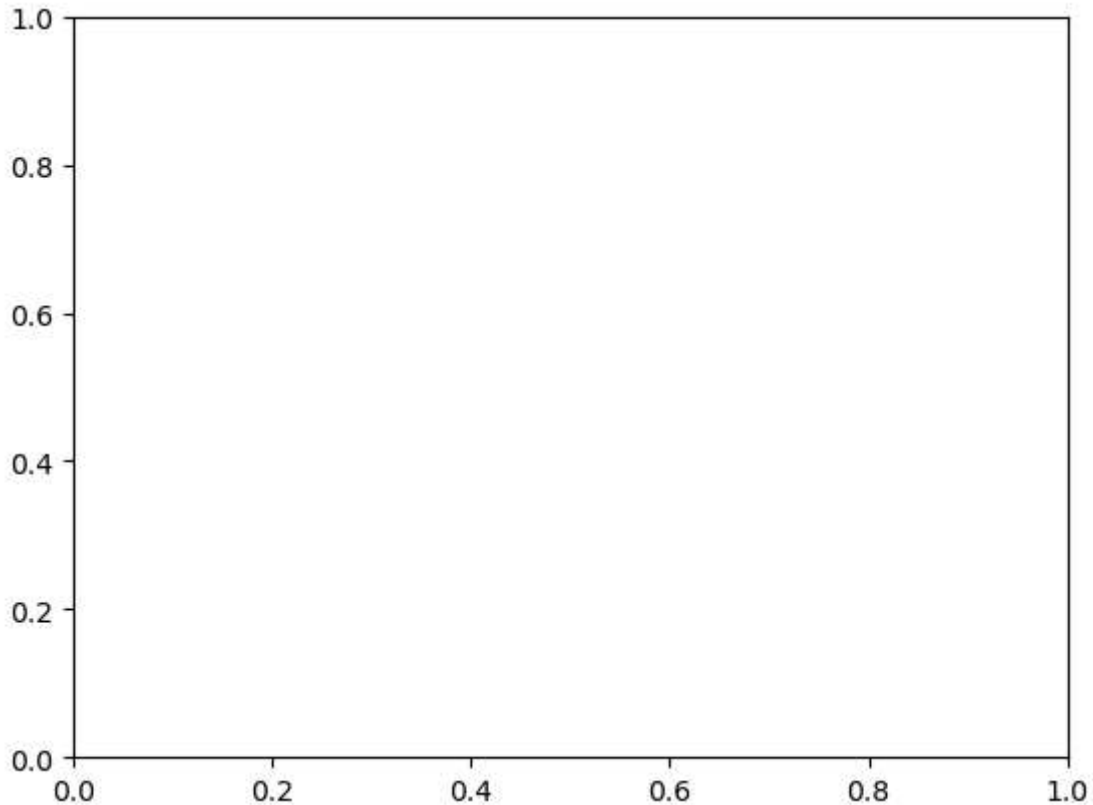
```

128 if _print_supported_values:
129     msg += f"; supported values are {'', '.join(map(repr, values))}"
--> 130 raise ValueError(msg)

```

ValueError: 'greens' is not a valid value for cmap; supported values are 'Accent', 'Accent_r', 'Blues', 'Blues_r', 'BrBG', 'BrBG_r', 'BuGn', 'BuGn_r', 'BuPu', 'BuPu_r', 'CMRmap', 'CMRmap_r', 'Dark2', 'Dark2_r', 'GnBu', 'GnBu_r', 'Greys', 'Greys_r', 'Greens', 'Greens_r', 'Greys', 'Greys_r', 'OrRd', 'OrRd_r', 'Oranges', 'Oranges_r', 'PRGn', 'PRGn_r', 'Paired', 'Paired_r', 'Pastel1', 'Pastel1_r', 'Pastel2', 'Pastel2_r', 'PiYG', 'PiYG_r', 'PuBu', 'PuBuGn', 'PuBuGn_r', 'PuBu_r', 'PuOr', 'PuOr_r', 'PuRd', 'PuRd_r', 'Purples', 'Purples_r', 'RdBu', 'RdBu_r', 'RdGy', 'RdGy_r', 'RdPu', 'RdPu_r', 'RdYlBu', 'RdYlBu_r', 'RdYlGn', 'RdYlGn_r', 'Reds', 'Reds_r', 'Set1', 'Set1_r', 'Set2', 'Set2_r', 'Set3', 'Set3_r', 'Spectral', 'Spectral_r', 'Wistia', 'Wistia_r', 'YlGn', 'YlGnBu', 'YlGnBu_r', 'YlGn_r', 'YlOrBr', 'YlOrBr_r', 'YlOrRd', 'YlOrRd_r', 'afmhot', 'afmhot_r', 'autumn', 'autumn_r', 'berlin', 'berlin_r', 'binary', 'binary_r', 'bone', 'bone_r', 'brg', 'brg_r', 'bwr', 'bwr_r', 'cividis', 'cividis_r', 'cool', 'cool_r', 'coolwarm', 'coolwarm_r', 'copper', 'copper_r', 'cubehelix', 'cubehelix_r', 'flag', 'flag_r', 'gist_earth', 'gist_earth_r', 'gist_gray', 'gist_gray_r', 'gist_grey', 'gist_grey_r', 'gist_heat', 'gist_heat_r', 'gist_ncar', 'gist_ncar_r', 'gist_rainbow', 'gist_rainbow_r', 'gist_stern', 'gist_stern_r', 'gist_yarg', 'gist_yarg_r', 'gist_yerg', 'gist_yerg_r', 'gnuplot', 'gnuplot2', 'gnuplot2_r', 'gnuplot_r', 'gray', 'gray_r', 'grey', 'grey_r', 'hot', 'hot_r', 'hsv', 'hsv_r', 'inferno', 'inferno_r', 'jet', 'jet_r', 'magma', 'magma_r', 'managua', 'managua_r', 'nipy_spectral', 'nipy_spectral_r', 'ocean', 'ocean_r', 'pink', 'pink_r', 'plasma', 'plasma_r', 'prism', 'prism_r', 'rainbow', 'rainbow_r', 'seismic', 'seismic_r', 'spring', 'spring_r', 'summer', 'summer_r', 'tab10', 'tab10_r', 'tab20', 'tab20_r', 'tab20b', 'tab20b_r', 'tab20c', 'tab20c_r', 'terrain', 'terrain_r', 'turbo', 'turbo_r', 'twilight',

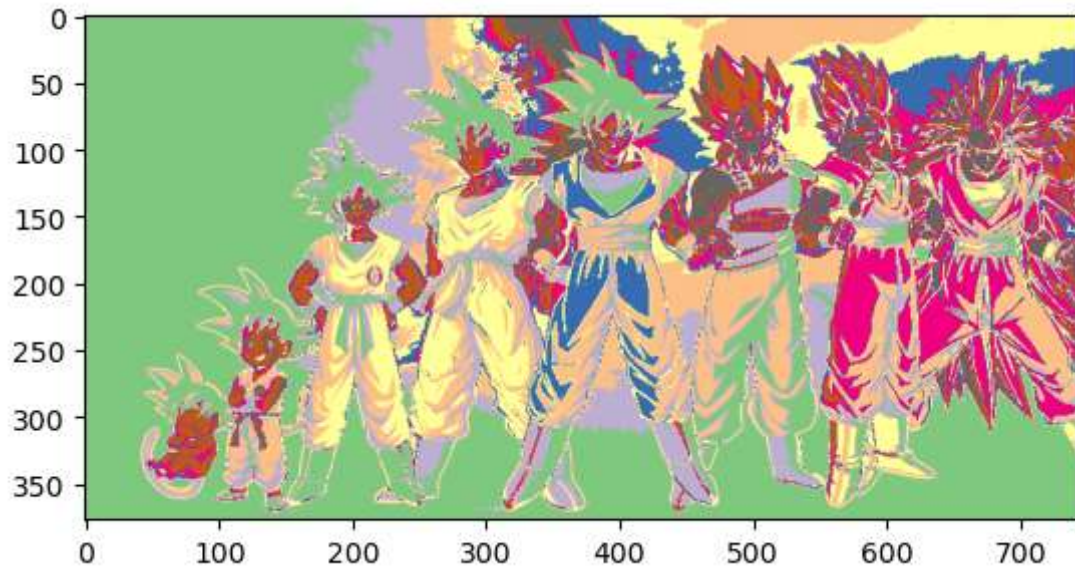
```
'twilight_r', 'twilight_shifted', 'twilight_shifted_r', 'vanimo', 'vanimo_r', 'viridis', 'viridis_r', 'winter', 'winter_r'
```



valid colours 'greens' is not a valid value for cmap; supported values are 'Accent', 'Accent_r', 'Blues', 'Blues_r', 'BrBG', 'BrBG_r', 'BuGn', 'BuGn_r', 'BuPu', 'BuPu_r', 'CMRmap', 'CMRmap_r', 'Dark2', 'Dark2_r', 'GnBu', 'GnBu_r', 'Grays', 'Grays_r', 'Greens', 'Greens_r', 'Greys', 'Greys_r', 'OrRd', 'OrRd_r', 'Oranges', 'Oranges_r', 'PRGn', 'PRGn_r', 'Paired', 'Paired_r', 'Pastel1', 'Pastel1_r', 'Pastel2', 'Pastel2_r', 'PiYG', 'PiYG_r', 'PuBu', 'PuBuGn', 'PuBuGn_r', 'PuBu_r', 'PuOr', 'PuOr_r', 'PuRd', 'PuRd_r', 'Purples', 'Purples_r', 'RdBu', 'RdBu_r', 'RdGy', 'RdGy_r', 'RdPu', 'RdPu_r', 'RdYlBu', 'RdYlBu_r', 'RdYlGn', 'RdYlGn_r', 'Reds', 'Reds_r', 'Set1', 'Set1_r', 'Set2', 'Set2_r', 'Set3', 'Set3_r', 'Spectral', 'Spectral_r', 'Wistia', 'Wistia_r', 'YlGn', 'YlGnBu', 'YlGnBu_r', 'YlGn_r', 'YlOrBr', 'YlOrBr_r', 'YlOrRd', 'YlOrRd_r', 'afmhot', 'afmhot_r', 'autumn', 'autumn_r', 'berlin', 'berlin_r', 'binary', 'binary_r', 'bone', 'bone_r', 'brg', 'brg_r', 'bwr', 'bwr_r', 'cividis', 'cividis_r', 'cool', 'cool_r', 'coolwarm', 'coolwarm_r', 'copper', 'copper_r', 'cubehelix', 'cubehelix_r', 'flag', 'flag_r', 'gist_earth', 'gist_earth_r', 'gist_gray', 'gist_gray_r', 'gist_grey', 'gist_grey_r', 'gist_heat', 'gist_heat_r', 'gist_ncar', 'gist_ncar_r', 'gist_rainbow', 'gist_rainbow_r', 'gist_stern', 'gist_stern_r', 'gist_yarg', 'gist_yarg_r', 'gist_yerg', 'gist_yerg_r', 'gnuplot', 'gnuplot2', 'gnuplot2_r', 'gnuplot_r', 'gray', 'gray_r', 'grey', 'grey_r', 'hot', 'hot_r', 'hsv', 'hsv_r', 'inferno', 'inferno_r', 'jet', 'jet_r', 'magma', 'magma_r', 'managua', 'managua_r', 'nipy_spectral', 'nipy_spectral_r', 'ocean', 'ocean_r', 'pink', 'pink_r', 'plasma', 'plasma_r', 'prism', 'prism_r', 'rainbow', 'rainbow_r', 'seismic', 'seismic_r', 'spring', 'spring_r', 'summer', 'summer_r', 'tab10', 'tab10_r', 'tab20', 'tab20_r', 'tab20b', 'tab20b_r', 'tab20c', 'tab20c_r', 'terrain', 'terrain_r', 'turbo', 'turbo_r', 'twilight', 'twilight_r', 'twilight_shifted', 'twilight_shifted_r', 'vanimo', 'vanimo_r', 'viridis', 'viridis_r', 'winter', 'winter_r'

```
In [37]: plt.imshow(image_gray, cmap = 'Accent')
```

```
Out[37]: <matplotlib.image.AxesImage at 0x14a3ce8efd0>
```



```
In [38]: plt.imshow(image)
```

```
Out[38]: <matplotlib.image.AxesImage at 0x14a3cee2c10>
```



```
In [40]: plt.imshow(fix_image) #####
```

```
Out[40]: <matplotlib.image.AxesImage at 0x14a3d0fa210>
```



```
In [41]: fix_image.shape
```

```
Out[41]: (377, 745, 3)
```

now we resize this image

```
In [45]: fix_image_1 = cv2.resize(fix_image,(188,362))
```

```
In [46]: fix_image_1
```

```

Out[46]: array([[ 12,  20,  59],
                [ 11,  19,  57],
                [ 11,  19,  56],
                ...,
                [ 53, 117, 189],
                [ 53, 120, 191],
                [ 54, 118, 190]],

                [[ 11,  19,  58],
                [ 11,  19,  57],
                [ 11,  19,  56],
                ...,
                [ 53, 117, 189],
                [ 52, 119, 190],
                [ 53, 120, 191]],

                [[ 10,  18,  58],
                [ 11,  19,  57],
                [ 11,  19,  55],
                ...,
                [ 53, 120, 191],
                [ 52, 119, 190],
                [ 53, 121, 192]],

                ...,

                [[  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0],
                ...,
                [  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0]],

                [[  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0],
                ...,
                [  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0]],

                [[  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0],
                ...,
                [  0,   0,   0],
                [  0,   0,   0],
                [  0,   0,   0]]], shape=(362, 188, 3), dtype=uint8)

```

```
In [47]: fix_image_1.shape
```

```
Out[47]: (362, 188, 3)
```

or we can also write a different code

```
In [48]: w_ratio = 0.5  
        h_ratio = 0.5
```

```
In [50]: fix_image_2 = cv2.resize(fix_image,(0,0), fix_image,w_ratio,h_ratio)
```

```
In [51]: fix_image_2
```

```

Out[51]: array([[ 12,  20,  59],
               [ 11,  19,  57],
               [ 11,  19,  56],
               ...,
               [ 53, 119, 190],
               [ 54, 120, 191],
               [ 53, 121, 192]],

            [[ 11,  19,  57],
             [ 11,  19,  56],
             [ 11,  19,  56],
             ...,
             [ 54, 122, 193],
             [ 56, 123, 194],
             [ 55, 124, 195]],

            [[ 11,  19,  56],
             [ 12,  20,  57],
             [ 11,  19,  55],
             ...,
             [ 57, 126, 196],
             [ 57, 125, 196],
             [ 59, 129, 199]],

            ...,

            [[  0,   0,   0],
             [  0,   0,   0],
             [  0,   0,   0],
             ...,
             [  0,   0,   0],
             [  0,   0,   0],
             [  0,   0,   0]],

            [[  0,   0,   0],
             [  0,   0,   0],
             [  0,   0,   0],
             ...,
             [  0,   0,   0],
             [  0,   0,   0],
             [  0,   0,   0]],

            [[  0,   0,   0],
             [  0,   0,   0],
             [  0,   0,   0],
             ...,
             [  0,   0,   0],
             [  0,   0,   0],
             [  0,   0,   0]]], shape=(94, 186, 3), dtype=uint8)

```

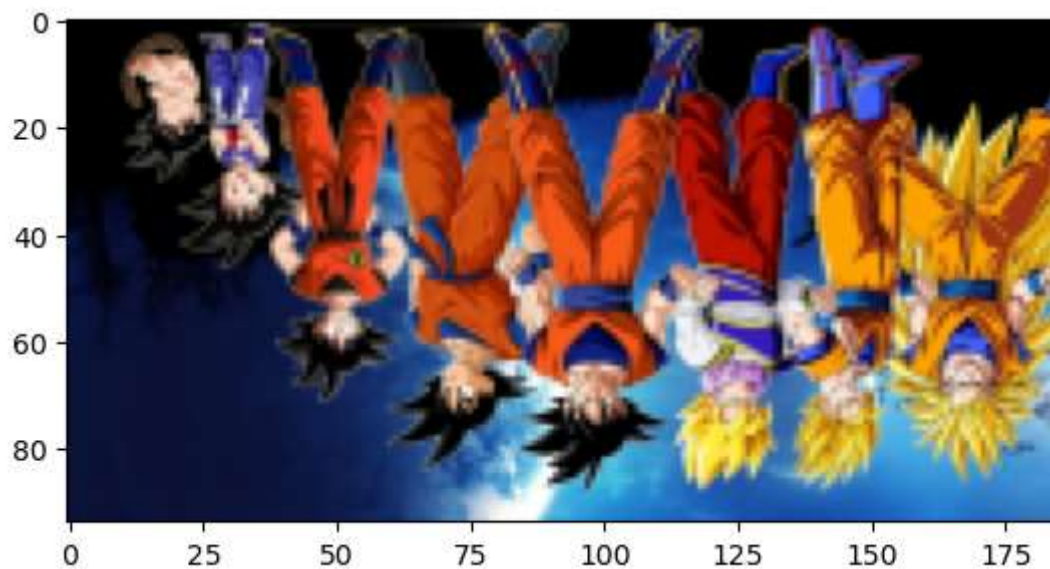
```
In [52]: plt.imshow(fix_image_2)
```

```
Out[52]: <matplotlib.image.AxesImage at 0x14a3d179d10>
```



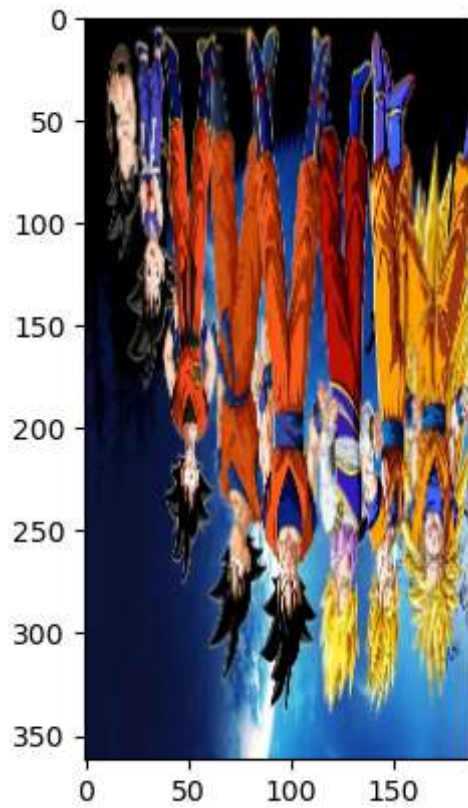
```
In [53]: image3 = cv2.flip(fix_image_2,0)  
plt.imshow(image3)
```

Out[53]: <matplotlib.image.AxesImage at 0x14a3d1f02d0>



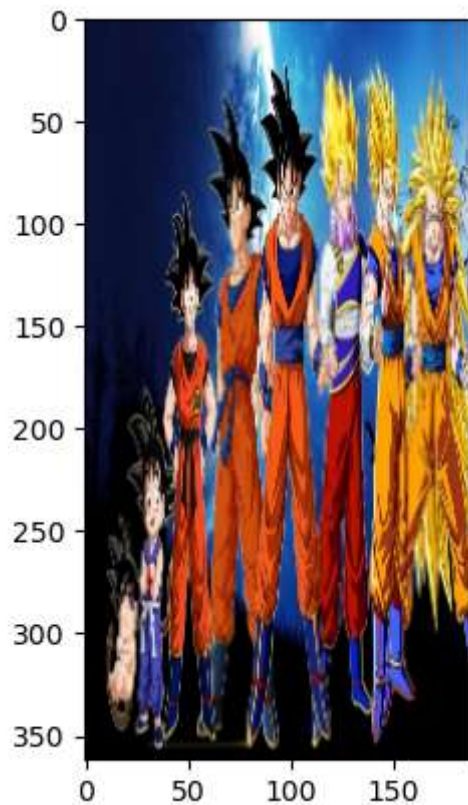
```
In [54]: image3 = cv2.flip(fix_image_1,0)  
plt.imshow(image3)
```

Out[54]: <matplotlib.image.AxesImage at 0x14a3cf12850>



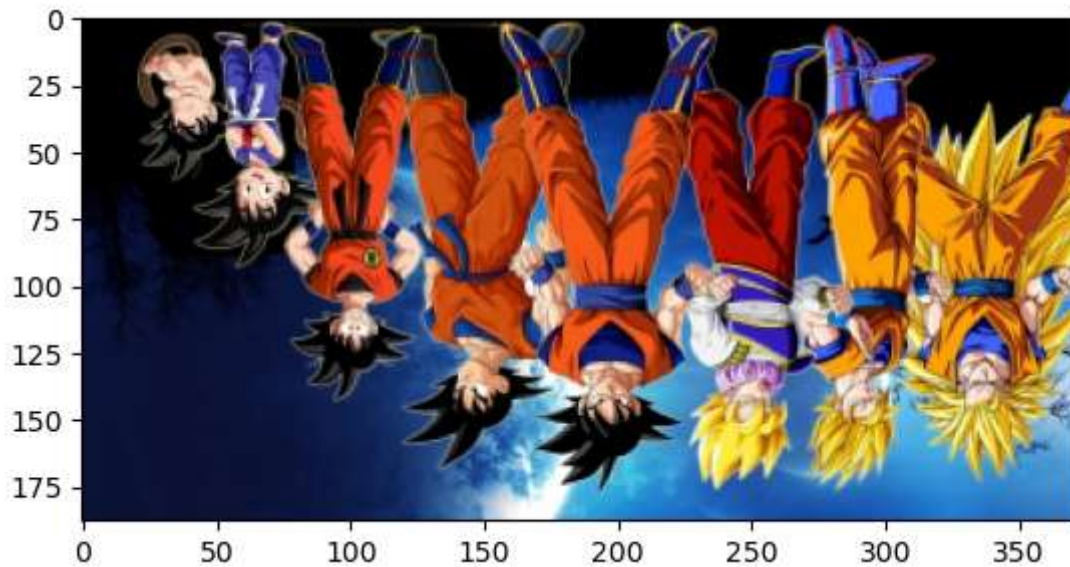
```
In [55]: plt.imshow(fix_image_1)
```

```
Out[55]: <matplotlib.image.AxesImage at 0x14a3cf90690>
```



```
In [56]: image3 = cv2.flip(fix_image,0)  
plt.imshow(image3)
```

Out[56]: <matplotlib.image.AxesImage at 0x14a3cfea490>



```
In [57]: image3 = cv2.flip(fix_image,1)  
plt.imshow(image3)
```

Out[57]: <matplotlib.image.AxesImage at 0x14a3d3f20d0>



```
In [58]: plt.imshow(fix_image)
```

Out[58]: <matplotlib.image.AxesImage at 0x14a3d471d10>



```
In [59]: image3 = cv2.flip(fix_image_2,1)
plt.imshow(image3)
```

Out[59]: <matplotlib.image.AxesImage at 0x14a3f559950>



```
In [60]: image3 = cv2.flip(fix_image_2,3)
plt.imshow(image3)
```

Out[60]: <matplotlib.image.AxesImage at 0x14a3f5b7ed0>



to save this image : 62 CELL

```
In [62]: cv2.imwrite("new_genai_image.jpg",image3)
```

```
Out[62]: True
```

```
In [64]: pwd #path location where files are stored
```

```
Out[64]: 'C:\\Users\\Abhilash Reddy'
```

to see where the image is stored go to the home page of jupyter notebook you will find images there

project 2 : image processing with open cv is done

```
In [ ]:
```